Ali Asadi

Interests

Technical lead and developer with a passion for novel computing technologies, fine-tuning algorithms, and craft high-performance software and tools. Key skills:

Programming Languages: `C`, `C++`, `Python`, **also familiar with** `Rust` and `Mojo`

HPC Tools and Extensions: `AVX`, `OpenMP`, `Kokkos`, `CUDA`, `HIP`, `MPI`, `Dask`, `Ray`

Scientific software development, debugging, profiling, runtime and compilers

Experience

Xanadu Quantum Technologies Sept. 2021 – Present
Performance Tech Lead and Sr. Quantum Software Developer *Toronto, Canada*

> I started my journey by crafting PL-Lightning and PL-Lightning-GPU: open-source `C++17/20` high-performance AVX, OpenMP, MPI, and GPU enabled state-vector Quantum simulators.

> I introduced distributed state-vector quantum simulation in PL, extended Python bindings, code sanitizers and software packaging, and enhanced the performance and scalability of PennyLane and Lightning simulators on HPC workstations.

> I contributed to the design and development of PL-Catalyst: an end-to-end JIT compiler and runtime targeting simulators and Quantum hardware hinge on QIR, LLVM/MLIR and JAX.

> As the lead developer of Catalyst Runtime, I crafted an extensible, thread-safe, multi-device QIR and QASM runtime with async-execution and dynamic plugin device supports.

> I am currently a performance technical lead, helping the software team to boost PennyLane + Lightning + Catalyst efficiency and scalability, develop new simulators, and ensure the performance of hybrid Quantum Machine Learning workflows backed by JAX, AutoGrad, TensorFlow and PyTorch.

Western University Sept. 2017 – Sept. 2021
Research Assistant *London, Canada*

> My Ph.D. thesis centered around crafting the fastest symbolic system solver based on RegularChain in `C` & `C++17`, packaged in BPASlib. I led the development of efficient algorithms for dense polynomial arithmetic over prime fields and big integers. I developed specialized, cache-friendly, and parallel arithmetic for dense and sparse multivariate polynomials, as well as the high-performance BPAS solver.

Maplesoft June 2020 – Sept. 2020
R&D Software Developer Intern *Waterloo, Canada*

> I contributed to the design and development of multivariate power series library in Maple 2021. I further enhanced the performance of basic arithmetic, boosting factorization by up to 500x.

Education

Ph.D. in Computer Science Western University, Canada
HPC in Scientific Computing *2017 – 2021*

M.Sc. in Computer Science Isfahan University of Technology, Iran
Cryptography and Coding Theory *2014 – 2017*

B.Sc. in Mathematics Isfahan University of Technology, Iran
Computational and Applied Mathematics *2010 – 2014*

Awards

- Mitacs Research and Development Scholarship for Internship at Maplesoft, 2020
- Western Graduate Research Scholarship for Ph.D. in Computer Science, 2017
- Ranked First among the M.Sc. students of Class 2014 in the department
- Ranked First among the B.Sc. students of Class 2010 in the department
- Ranked First in Mathematics and Computing University Competition, 2011

LinkedIn | Publications | GitHub